Telicity In Aspect Acquisition: Evidence From Mandarin Children

Abstract

This study investigates Mandarin-speaking children's acquisition of verb-framed telicity encoded by resultative compounds and completive particles, two mechanisms operating independent of object quantization. Using picture-selection tasks, I observed a developmental dissociation: children showed stable performance with completive particles from age 3, whereas comprehension of resultative compounds improved gradually with age. The asymmetry suggests distinct developmental pathways for these two constructions. The observed asymmetry between resultative compounds and completive particles is more readily captured under an analysis that locates telicity within verb phrase (e.g. Ramchand, 2008). Children's steady performance with completive particles may reflect an early sensitivity to event representation, while the protracted refinement of resultative verb compounds suggests that the mapping between syntax-semantics form and event representation gradually becomes more precise. This work underscores that the acquisition of telicity is a parametric process, shaping and being shaped by event cognition.

Keywords: telicity, child acquisition, event structure, syntax-semantics interface

1. Introduction

1.1 Telicity in aspect

Aspect encompasses two distinct but interacting dimensions (Smith, 1997). Viewpoint (or grammatical) aspect concerns a speaker's perspective on an event, such as presenting it as a whole (perfective) or from within (imperfective). In contrast, situational aspect refers to the inherent temporal contour of an event as depicted by a predicate, categorizing events into types such as states, activities, accomplishments, and achievements. A central feature within this classification is telicity, the property of a predicate describing an event with an inherent endpoint. Critically, telicity is a property of the linguistic description, not of the extralinguistic event; the same real-world happening

can be framed as telic (build a house) or atelic (build). This study focuses on the acquisition of telicity in Mandarin, a core component of situational aspect.

Vendler (1957) classified verbs into stative verbs (static and continuous status, like *know*, *love*), activity verbs (continuous action without a natural end, like *walk*, *ride*), accomplishment verbs (continuous action with a natural end, like *build* and *eat*), and achievement verbs (action with a single moment and a natural end, like *reach*, *end*). Telicity distinguishes accomplishment and achievement verbs from activity and stative verbs. The standard diagnostic uses *in*-adverbials (compatible with telic predicates) versus *for*-adverbials (compatible with atelic predicates), as shown in (1) and (2).

(1) He built a model plane in/*for a week.

(2) John walked for/*in an hour.

While Vendler's classification focuses primarily on verbs, telicity is a property of the entire predicate, not the verb alone (Filip, 1999). For accomplishment predicates with incremental themes, the presence of a quantized object may shift a predicate from atelic to telic; this quantization effect does not apply to activity or achievement verbs. A noun phrase is quantized if it refers to a discrete unit or specified quantity (e.g. an apple, two cups of water), and cumulative if it refers to an unspecified amount or mass (e.g. apples, water). Under Filip's (1999) analysis, quantized objects yield telic interpretations at the predicate level, while cumulative objects yield atelic interpretations. Thus, verbs like *build* or *eat* are unspecified for telicity: they are telic with quantized objects (e.g. build a house, eat an apple) and atelic with cumulative ones (e.g. build houses, eat apples).

For verbs whose event progression is measured by their internal arguments (i.e. incremental theme verbs like *eat* or *build*), the part-structure of the object maps onto the part-structure of the event (Krifka, 1992). In (3), the event of eating progresses as the apple is consumed; once the apple is fully consumed, the event reaches its endpoint. A goal phrase can further specify an evental endpoint, as in (4).

(3) I have eaten an apple.

(4) I have eaten an apple to the core.

For route verbs like *walk* or *climb*, the object (e.g. the trail, the ladder) does not inherently specify an

evental endpoint. Telicity is indeterminate unless a goal phrase is added, as shown in (5)-(8).

(5) I walked the trail in/for an hour.

(6) I climbed the ladder in/for two minutes.

(7) I walked the trail to its end in/*for an hour.

(8) I climbed the ladder to the top in/*for two minutes.

Cross-linguistically, telicity is primarily determined through two sources: the quantization of a nominal argument as an incremental theme and the presence of an overt resultative phrase (ResP)[1]. However, languages differ significantly in how these sources are morphosyntactically instantiated. For instance, while English often correlates telicity with the formal properties of the object DP (e.g. quantized versus bare nominals), this correlation is not always sufficient (Filip, 2004). Phrases like *carried three pears* and *stirred the soup* are atelic despite having quantized or definite objects, because the verbs are not incremental; conversely, *found gold* is telic even with a bare object, because the finding event is inherently atomic (Filip, 2004). Similarly, Slavic languages exhibit rich systems of verb prefixes, but these prefixes are best characterized as lexical-derivational morphemes with diverse semantic functions (e.g. direction, quantity, manner), rather than as systematic grammatical markers of telicity (Filip, 2004). The theoretical significance of this cross-linguistic variation lies in how it is captured by competing syntactic analyses, to which we now turn.

1.2 Telicity encoding in Mandarin Chinese

The cross-linguistic variation in telicity encoding raises a critical question: how does a language encode telicity if it does not rely on object quantization as its primary mechanism, and instead grammaticalizes telicity through verb-framed syntax? How do children acquire such language? Under Ramchand's (2008) account, Slavic prefixed verbs, English verb-particle constructions, and Mandarin resultative verb compounds all realize the abstract resultative functional head, differing only in morphological realization: Slavic languages fuse result via head-movement into verbal stems, while Mandarin realizes result as a separate verbal element within resultative compounds (Basciano, 2010). Unlike English, where object quantization exerts central aspectual effects, Mandarin is argued

[1] See relevant discussion in section 1.2.

to deploy two related verb-framed strategies (Wang, 2024): resultative verb compounds (Deng, 2019; Tay, 2026) and completive particles (Gu, 2023; Xiong & Li, 2026). These mechanisms operate independently of object quantization, allowing flexible interaction between verb semantics, event structure, and syntactic composition (Wang, 2024). The heavy reliance on verb-framed syntax via ResP shows that while abstract semantic features such as [±telic] may be universal, the syntactic means to value these features are parametric. This motivates shifting inquiry away from whether telicity is universally tied to quantization, toward mapping the full parametric space for telicity-encoding mechanisms across languages.

One of the most salient features of Mandarin telicity is the role of resultative verb compounds, which consist of a main verb followed by a secondary predicate indicating the resultative state (Deng, 2019; Tay, 2026). The construction explicitly encodes telicity of an event. In (9), the resultative verb compound imposes a telic reading, despite the object *衣服* /yī fu/ 'the clothes' being a bare noun with no inherent quantization. This phenomenon contrasts sharply with English, where a quantized object (e.g. 'the clothes') would typically be required for a telic interpretation. Crucially, a resultative verb compound does not necessarily confer definiteness or specificity to an object, meaning that even non-referential nouns can appear in telic events (Hu, 2018). Thus, Mandarin relies more heavily on lexical-semantic composition (i.e. the verb-result pairing) rather than syntactic quantization to determine telicity.

(9) 娜娜 在 三分钟 内 洗 干净 衣服
nà nà zài sān fēn zhōng nèi xǐ gān jìng yī fu
Nana in 3-minute in wash clean-RES clothes
'Nana washed the clothes clean in three minutes.'

(Wang, 2024)

A second key mechanism is the use of completive particles, such as *完* /wán/ 'finish', *掉* /diào/ 'off/away', *到* /dào/ 'reach/onto', *好*/hǎo/ 'ready/done', and *住* /zhù/ 'secured' (Gu, 2023; Xiong & Li, 2026). Unlike Slavic perfective prefixes, these particles do not carry quantificational force and instead grammaticalize mere telic status without implying a change of state.[2] In (10), the particle *完*

[2] Gu (2023) further distinguishes between termination (event cessation regardless of whether a resultative state is achieved) and culmination (attainment of a resultative state) in the analysis of completive particles *完* /wán/ and *掉* /diào/. I follow the standard practice of grouping completive particles as a class, without experimentally committing to the termination/culmination distinction.

/wán/ signals that the event is telic, but the object *饼干*/bǐng gān/ 'biscuits' remains non-quantized---there is no requirement that all biscuits are consumed. Completive particles differ from resultative verb compounds, which as a class entail an actual resultative state. The optionality of object quantization raises questions for strictly quantization-based accounts (Borer, 2005), although more research is needed to explore whether covert or context-dependent quantization exists. Instead, Mandarin suggests that telicity can be signaled by grammaticalized particles or lexical compounding, independently of nominal reference.

(10) 弟弟 吃完 饼干
dìdi chī wán bǐng gān
younger brother eat-COMPL biscuit
'The younger brother finished eating biscuits.'

A useful diagnostic distinguishes resultative verb compounds and completive particles. A resultative verb compound entails a specific resultative state predicated of the object, paraphrasable as "make/become+adjectival compound". A completive particle, by contrast, only entails the status of being telic, paraphrasable as "finish V_{ing}", and tolerates continuation modifiers like "but not completely" (e.g. finished the eating event but did not completely eat all the biscuits), which any resultative verb compound rejects.

A further complication arises with verbs like *吃* /chī/ 'eat', which in Mandarin exhibit aspectual variability even with quantized objects. (11) can also permit atelic interpretations, as "the boy could eat an apple and may not finish eating it". This ambiguity suggests that object quantization does not function as a strict determinant of telicity in Mandarin. Instead, the primary cue for telicity is the presence of an explicit result phrase (resultative verb compounds or completive particle), while bare quantized objects may or may not enforce a telic event interpretation (Wang, 2024).

(11) 男孩 吃 一个 苹果
nán hái chī yí ge píng guǒ
boy eat CL apple
'The boy ate an apple.'

These findings have significant theoretical implications. While Borer's exo-skeletal model (2005)

struggles to explain why telicity reliably emerges in Mandarin even with non-quantized objects (e.g. 洗干净衣服 /xǐ gān jìng yī fu/ ‘wash-clean-clothes’), Ramchand’s (2008) first-phase syntax captures this by positing that telicity is fundamentally a property of ResP (see **Figure 1a**). Under this view, the resultative complement (e.g. 干净 /gān jìng/ ‘clean’) or the completive particle (e.g. 完 /wán/ ‘finish’) is base-generated in Res, thereby syntactically constituting a telic event independently of the referential properties of the object DP. The object (e.g. 衣服 /yī fu/ ‘clothes’) is treated as the holder of the resultative state, namely [Spec, ResP], not the measure of the event. *干净* /gān jìng/ ‘clean’ moves from Res to adjoin to the verb *洗* /xǐ/ ‘wash’ in V, forming the compound *洗干净* /xǐ gān jìng/ ‘wash-clean’. In comparison, completive particles (e.g. 完 /wán/ ‘finish’ in 吃完 /chī wán/ ‘finish eating’) are also base-generated in Res but lack a state-change implication. They similarly undergo head movement to V. However, unlike resultative complements, they do not entail a specific resultative state (i.e. no equivalent to a resultative secondary predicate like *clean* in **Figure 1b**). This syntactic absence corresponds to their purely terminative semantic function. This configuration provides a principled explanation for verb-framed telicity[3] in Mandarin, shifting the theoretical burden from object quantization to the syntactic construction of resultativity.

(a) (b)

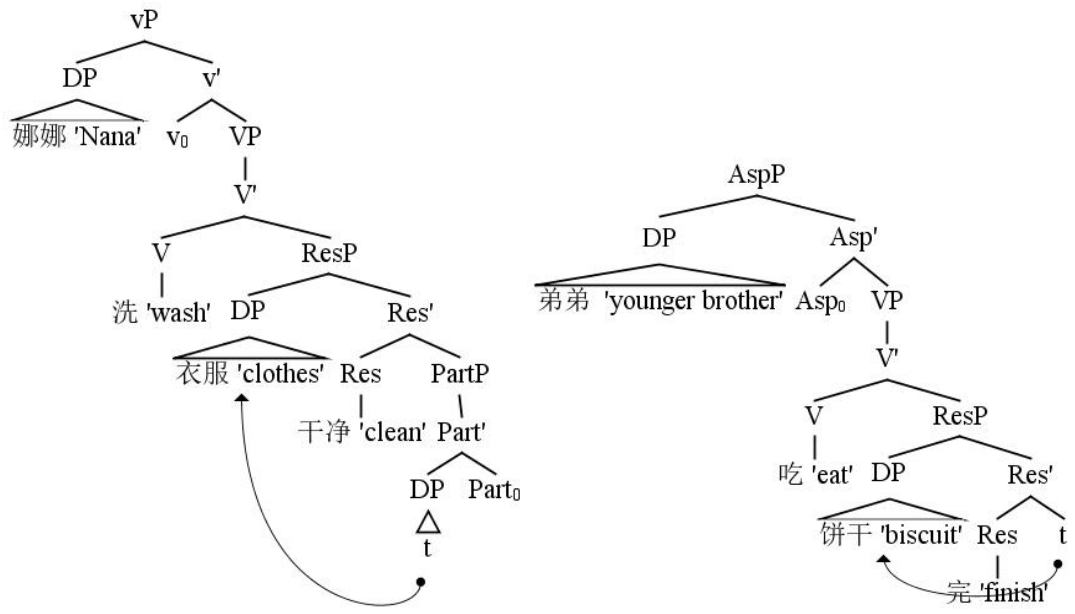


Figure 1. Syntactic structure of verb-framed telicity in Mandarin (adapted from Svenonius, 2005)

[3] Here, verb-framed telicity refers specifically to a typological parameter in which the primary syntactic projection of telicity originates from within the verb phrase (i.e. a ResP), independently of nominal quantization. The notion is distinct from the typology of motion-event encoding (i.e. verb-framed versus satellite-framed) in cognitive linguistics.

However, this model still faces challenges in distinguishing between resultative verb compounds and completive particles, as both are base-generated in the same Res head position and undergo identical head movement to combine with the main verb. Syntactically, this identical merge-and-move operation makes their structural representations nearly indistinguishable at the phrase-structure level. The semantic difference---state change versus mere evental termination (particles)---is not overtly marked in syntax but depends on lexical semantics (e.g. 干净 /gān jìng/ 'clean' implying a state versus 完 /wán/ 'finish' lacking it), forcing reliance on interpretive rather than structural cues for differentiation (Ramchand & Svenonius, 2002; Svenonius, 2005). Future research can further investigate how the lexical-aspectual system interacts with syntax to produce such flexible telicity encoding, particularly in comparison with other isolating languages that lack inflectional morphology.

1.3 Child acquisition of telicity

The acquisition of telicity engages a complex interface between linguistic structure and cognitive development, presenting children with the fundamental challenge of identifying which cues in their target language reliably signal telicity. While cross-linguistic evidence reveals that children are sensitive to a range of potential cues (including lexical semantics, syntactic frames, and pragmatic context), the weight assigned to each cue varies dramatically based on the structural properties of the language being acquired. This dependency makes the acquisition of Mandarin telicity a particularly compelling and theoretically informative case.

Crucially, Mandarin-acquiring children can't rely on the nominal quantization strategies that are primary in languages like English. Instead, they must master a system where telicity is predominantly encoded through verb-framed syntax. This typological difference predicts a distinct developmental pathway, requiring children to employ syntactic bootstrapping, namely the ability to use structural information to infer event semantics. Supporting this, Lu and Lee's (2019) work on Mandarin provided clear evidence for this capacity. Children were presented with sentences containing dual-category verbs (both unergative and unaccusative) like *run*. In the preverbal NP structure (e.g. the little pig ran just now), children accepted both telic and atelic interpretations. However, when the same verb appeared in the postverbal NP structure (e.g. there escaped/ran a little pig), they strongly preferred the telic interpretation, demonstrating their use of syntactic position as a decisive cue to

event telicity.

This syntactic bootstrapping capacity operates alongside children's emerging knowledge of telicity-encoding morphology in Mandarin. Two recent experimental studies examined the acquisition of resultative verb compounds. Deng (2019) analyzed longitudinal data from children aged 1;6-2;6 and conducted a novel-verb experiment with 2;9- to 4;0-year-olds. Children were able to decompose resultative verb compounds into their component verbs (e.g. recognizing that "wipe-clean" consists of an action "wipe" and a result state "clean"). Their understanding of event structure was acquired gradually rather than instantaneously: directional (e.g. climb up) and completive (e.g. eat up) verb compounds emerged earlier and were more compositional than result-state compounds (e.g. wipe-clean). Only 21% of children demonstrated full ability to both comprehend and produce novel verb compounds by age 4;0. Hsu et al. (2019) focused on the infixation of 不 /bu/ 'not able' and 得 /de/ 'be able' between the two verbs of a resultative verb compound (e.g. 拿不到 /ná bù dào/ 'can't reach' versus 拿得到 /ná de dào/ 'can reach') as a diagnostic of abstract compound representation. Despite the rarity of such infixed forms in child-directed input (less than 0.7% of input utterances), 3- to 5-year-olds in an elicitation task performed exceptionally well when provided with discourse-appropriate prompts, indicating early productivity with the compound schema. These findings collectively show that children acquire resultative verb compounds as a productive schema well before age 6, but they leave open how children integrate resultative verb compounds and completive particles into their interpretive system of telicity, and whether these two mechanisms follow the same developmental timetable.

This pattern of early sensitivity to verb-framed cues in Mandarin aligns with broader cross-linguistic evidence that transparent syntactic cues are acquired early. For instance, Wagner (2010) found that English-speaking children, even with novel verbs, associated transitive frames (e.g. the boy dracked the X) with telic events in match-to-sample tasks. In contrast, the mastery of morpho-syntactic devices that encode telicity, including aspectual particles with a lexical basis such as English *up*, often extends into middle childhood. Evidence on the developmental timeline is mixed: while Jeschull (2007) found that English-speaking children under 6 struggled to distinguish particle verbs (e.g. eat up) from simplex verbs (e.g. eat), Van Hout (1998) reported that 5-year-olds could make this distinction.

Beyond pure grammatical factors, research on pragmatic influences showed that children's telicity judgments were affected by discourse framing (Anderson, 2017). In a truth-value judgment task, children were more likely to reject a sentence like "she cut the paper" after seeing an incomplete event if a completed scenario was shown first, indicating their interpretations were more context-dependent than adults'. This slow maturation underscores the theoretical importance of investigating the 3- to 6-year-old age range in Mandarin, a period likely critical for consolidating the abstract knowledge required to deploy resultative compounds and completive particles as independent telicity markers.

Furthermore, findings from atypical and bilingual populations highlight the cognitive dimensions of this acquisition process. The selective deficit in German-speaking children with specific language impairment, who consistently failed to reject telic verbs like *aufmachen* 'open' for incomplete events in yes/no questions, contrasts with their preserved ability to comprehend compositional markers like *aufessen* 'eat up' (Penner et al., 2003; cited by Arunachalam & Syrett, 2018). This finding suggests distinct processing mechanisms for different aspectual cues. Conversely, the metalinguistic advantages observed in balanced bilinguals (compared to dominant bilinguals) during picture description tasks (Dosi, 2019)---where they more accurately selected grammatical aspect matching ongoing versus completed events---point to the role of cognitive flexibility. Together, these strands of research converge on a central theme: the acquisition of telicity is an adaptive, resource-sensitive process.[4] The present study leverages this foundational understanding by systematically probing how Mandarin children navigate this learning problem, thereby illuminating the interplay between universal cognitive capacity for understanding event structure and language-specific encoding.

2. The Present Study

The current study investigated how Mandarin-speaking children acquire how telicity is encoded, a crucial yet underexplored dimension in child acquisition research. Two research questions were formulated:

[4] For more literature related to the acquisition of telicity, please see Hodgson (2009 & 2010) and Stoicescu & Dressler (2022).

- Whether, and at what age, Mandarin children recognize the independent coding effect of result complements on telicity?
- Whether, and at what age, Mandarin children differentiate the effect between completive particles and bare verbs on telicity?

On the one hand, the present study is motivated by a critical gap at the intersection of syntax-semantics theory and language acquisition. While cross-linguistic research has firmly established that children can acquire telicity through object quantization or perfective morphology, it remains unknown how they acquire a system like Mandarin, where telicity is primarily encoded by verb-framed mechanisms that operate independently of these common cues. This typological profile makes Mandarin an interesting ground for the long-standing theoretical debate between Borer's (2005) exo-skeletal model, which posits object quantization as fundamental, and Ramchand's (2008) first-phase syntax, which grants ResP this central role. By examining whether children treat resultative compounds and completive particles as reliable telicity markers even without overtly quantized objects, this study assessed whether children are sensitive to verb-framed cues to telicity independently of overt object quantization. Rather than adjudicating competing theories of adult grammar, the goal is to examine whether the developmental data are more naturally accommodated by an analysis that locates telicity in the verb phrase versus one that derives it from nominal quantization.

To isolate the acquisition of these specific verb-framed mechanisms, the study focused on resultative complements and completive particles while controlling for other potential cues. Most notably, viewpoint-aspect markers, including 了/le/, 着/zhe/, 过/guò/, and 在/zài/, were excluded from the experimental stimuli. This exclusion is methodologically and theoretically crucial. While these markers may interact with telicity, they do not inherently encode telicity. Verb suffix 了/le/ functions as a perfective marker, whereas sentence-final 了/le/ operates as a current relevance marker. Their semantic interpretation is heavily context-dependent (Wu, 2022). Viewpoint aspect is logically separable from event telicity. Children as young as 4-6 years old demonstrate early sensitivity to probabilistic associations between viewpoint aspect and lexical aktionsart (Li & Bowerman, 1998). Specifically, children tend to pair the imperfective marker 在/zài/ with atelic activity verbs (e.g. run), and associate the verb-suffix perfective 了/le/ more strongly with telic accomplishment/achievement

verbs (e.g. build). Upon encountering verb-suffix 了 /le/, children are more prone to infer event completion when the predicate is lexically telic. These findings show children draw on lexical aspect as a bootstrapping cue for interpreting aspect markers, shaping their construal of an event as ongoing or completed. The probabilistic association between aspect and telicity motivates my choice of controlled stimuli.

On the other hand, the developmental journey of acquiring the telic/atelic distinction probes an issue in developmental psychology: how does children's early non-linguistic sensitivity to event boundary become integrated with the language-specific syntactic structures that encode telicity in their target language? While infants demonstrate a non-linguistic sensitivity to event boundary (Baldwin et al., 2001; Loucks & Meltzoff, 2013; Sonne et al., 2016), the transition to treating these cognitive categories as grammatically encoded features represents a critical step in experimental acquisition. This process is not a simple matter of labeling independently established conceptual categories. Rather, it involves discovering which linguistic forms in the input (whether nominal, verbal, or morphological) reliably signal telicity in a target language. The case of Mandarin is theoretically pivotal here because it challenges a potential universal default. One might hypothesize that a telic event, with its salient event boundary, is the cognitively privileged or unmarked interpretation for children. Here, "unmarked interpretation" does not refer to a syntactically overt default, but rather to the possibility that children may initially favor telic interpretations of described events, reflecting a general attentional preference for event endpoints that has been observed in non-linguistic event perception tasks with infants (Baldwin et al., 2001; Loucks & Meltzoff, 2013). This tendency may serve as a starting point for acquiring language-specific aspectual distinctions, rather than a fixed bias that overrides structural input. Therefore, the current study examined whether this proposed preference for telic event interpretation is robust enough to influence early syntax-semantics organization, even when the target language provides structural cues that do not consistently align with it.

Drawing on insights from both Ramchand's (2008) first-phase syntax and Borer's (2005) quantization-based model, two hypotheses were proposed corresponding to the research questions. For question one (result complements), if children systematically assign telic interpretations in the presence of resultative morphology even without overt quantized objects, this would indicate

sensitivity to morphosyntactic cues to telicity that are independent of object quantification. It is a pattern more consistent with an analysis that locates telicity in the resultative structure itself (Ramchand, 2008). A significant age-by-condition interaction would emerge, with accuracy on resultative compounds improving with age, while bare verbs remained stable across age groups. The alternative hypothesis should observe no such interaction, which would suggest that resultative morphology does not function as an independent telicity marker for children.

For question two (completive particles), two possible outcome patterns were considered. If completive particles and resultative compounds were to show parallel developmental trajectories, this would suggest that both constructions rely on a common underlying source of telicity, potentially compatible with accounts that derive telicity from a uniform mechanism such as object-related quantization. By contrast, if the two constructions were to diverge developmentally, this would indicate that different carriers of telicity are not acquired uniformly. In particular, a developmental asymmetry would suggest that telicity falls short of being reduced to a single source such as object-related quantization, and is more consistently captured if distinct mechanisms within the verb phrase contribute independently to telicity.

3. Methodology

3.1 Participants

The participants were recruited via convenience sampling. Parental informed consent was obtained for all the participants prior to their involvement in the study. 120 monolingual Mandarin-speaking children were stratified into 4 age groups: 3-, 4-, 5-, and 6-year-olds (30 participants per group). A separate cohort of 120 children with the same age distribution participated in experiment 2 to avoid carryover effects. All children were typically developing with no reported language or cognitive impairments. The participants were excluded if they scored below 80% accuracy in the filler condition. The exclusion criterion was formulated by referring to other experimental research (Pablos Robles et al., 2025; Solaimani, 2024). Applying this criterion resulted in the exclusion of data from 4 children in experiment 1 and 5 children in experiment 2. These excluded participants were evenly distributed across age groups, and their data were excluded from all analyses. The final analyzed sample thus

consisted of 116 children for experiment 1 and 115 children for experiment 2.

15 adults participated in the pilot study for both experiments, and a near-ceiling score was observed for both the resultative condition (mean=97.3%, SD=2.1) and the completive particle condition (mean=96.8%, SD=2.4). Hence, I did not recruit additional adult participants for formal experiments.

3.2 Materials

I employed a picture selection task as a children-friendly approach to examine learners' syntax-semantics acquisition of event structure. The test pictures were generated by the artificial intelligence tool OpenAI. Experiment 1 tested the role of result complements in telicity, featuring 15 scenarios per condition: bare verb condition (atelic), result complement condition (telic), and fillers. Each scenario was paired with three pictures: one correct, one incorrect, and one distractive. The decision to use 15 items per condition (45 trials in total) was a deliberate balance between statistical robustness and ecological validity for child participants. In a pilot study that employed 20-30 items per condition, I observed a clear decline in attention and task engagement in the later stages of the longer pilot sessions, leading to unreliable data. This trade-off is a well-recognized challenge in developmental psycholinguistics, consistent with established practices in the field. For instance, Köder and Falkum (2020) studied child metaphor comprehension in a picture-selection task, which used a small number of trials (20 trials in total). The presentation order of pictures across conditions and the set order of answers were pseudo-randomized. Take the bare verb condition as an example, (12) shows a full scenario presented to the participants (see **Figure 2-4**). Here, **Figure 2** should be the relatively best option if the sentence is presented with a resultative verb compound (i.e. 切开 /qiē-kāi/ 'cut-open'). By contrast, for the bare verb stimulus in (12), **Figure 3** constitutes the expected choice.

(12)

厨房	有	一	套	餐具	小明	切	面包
chú fáng	yǒu	yí	tào	cān jù	xiǎo míng	qiē	miàn bāo
kitchen	have	one	CL	tableware	Xiǎomíng	cut	bread
现在	的	场景	是	什么样	的	呢	
xiàn zài	de	chǎng jǐng	shì	shén me yàng	de	ne	
now	MOD	scene	COP	what	MOD	Q	

'Xiaoming has a set of tableware, and he cuts the bread. Which is the right picture?'



Figure 2. Picture A (incorrect): The bread is completely sliced.



Figure 3. Picture B (correct): The action of slicing the bread is in progress.



Figure 4. Picture C (interference): The action of slicing the orange is in progress.

Filler trials were incorporated to prevent the participants from developing a response strategy specific to telicity judgments and to monitor their general attention and comprehension. These trials depict states (e.g. the ground is wet) or simple events that are semantically distant from the telicity manipulation. For instance, a filler scenario describes a weather condition without an agent, and the picture selection task requires identifying the matching state (e.g. wet ground) while rejecting both a semantically opposite distractor (e.g. dry ground) and an unrelated interference (e.g. wet table).

Experiment 2 examined completive particles (e.g. *完* /wán/, *掉*/diào/, *到*/dào/, *好*/hǎo/, and *住*/zhù/) versus bare verbs, with 15 scenarios per condition: bare verb condition (atelic), completive particle condition (telic), and fillers (which resemble the design in experiment 1). Each scenario was paired with three pictures: one correct, one incorrect, and one distractive. The presentation order of pictures across conditions and the set order of answers were pseudo-randomized. Take the completive particle condition as an example, (13) shows a full scenario presented to the participants (see **Figure 5-7**). **Figure 6** should be the relatively best option.

(13)	小明	上	一	节	体育课	他	抓	住	蓝球
	xiǎo míng	shàng	yì	jié	tǐ yù kè	tā	zhuā	zhù	lán qiú
	Xiǎomíng	attend	one	CL	PE class	3SG	seize	COMPL	basketball
	现在	的	场景	是	什么样	的	呢		
	xiàn zài	de	chǎng jǐng	shì	shén me yàng	de	ne		
	now	MOD	scene	COP	what	MOD	Q		

'Xiaoming brings several classmates, and he catches the basketball. Which is the right picture?'



Figure 5. Picture A (incorrect): Xiaoming manages to catch the basketball but does not fully grab it.



Figure 6. Picture B (correct): Xiaoming successfully catches the basketball.



Figure 7. Picture C (interference): Xiaoming successfully catches the football.

Both experiments selected the 2500 most frequently used Chinese characters from BLCU Corpus Center (BCC) and a list of commonly used modern Chinese words (National Chinese Language Commission, 1988). This measure ensured that the stimuli were age-appropriate for 3-6-year-old children, preventing task performance from being confounded by unfamiliar vocabulary. In all stimuli, V+NP was set to follow a pattern of one character plus two characters (e.g. 折/zhé/ 'fold'+纸船/zhǐ chuán/ 'paper boat'). Both resultative verb compounds and completive particles should present as a single character to follow a verb. To eliminate lexical aktionsart and object quantization as confounding sources of telicity, all test items draw on inherently atelic activity verbs, where object quantization cannot produce telic readings in the first place. As a result, any telic interpretation must

be derived from verb-framed mechanisms, namely resultative complements or completive particles. This choice could prevent the independent effects of verb class on children's telicity judgments. An acceptability rating task (1=completely unnatural, 4=moderately acceptable, 7=fully natural) was administered offline with 15 native Mandarin adults for all critical experimental items. The items yielded a mean rating of 6.61 (SD = 0.35). The pilot test with 10 children confirmed the comprehensibility of stimuli, as they could accurately describe the intended event depicted in each image by using their own words. See full experimental stimuli in supplementary materials.

A potential limitation of the design is the absence of strict minimal pairs across conditions. A possible concern is that children's responses may be driven by their lexical knowledge of the resultative component itself rather than their understanding of its telicity-marking function. However, if children's success was solely due to recognizing this factor, performance should be uniformly high across age groups. The later reported age-by-condition interaction suggests that lexical recognition alone can't account for the developmental pattern. Although the design could be optimized, this alternative explanation should be denied.

3.3 Procedure

In both experiments, participants were tested individually in a quiet classroom setting to minimize distractions and maintain focus. The participants responded by using a paper-and-pencil answer sheet. Audio stimuli were prerecorded and spoken at a uniform pace to ensure consistent presentation across participants. They described the target event for each trial. The recordings were produced with natural prosody; no artificial pause or prosodic break was inserted between the verb and the following resultative complement or completive particle, thereby presenting each verb-framed structure as an integrated phonological unit. These sentences were played aloud while a PowerPoint slide simultaneously displayed three images corresponding to the atelic, telic, and filler interpretations of an event. The positions of images were randomized across trials to prevent response strategies. The participants had 20 seconds per trial to listen, view the images, and circle their selection on the answer sheet.

Prior to the main trials, children completed a training phase consisting of 12 practice trials (4 for each

condition: atelic, telic, and filler), designed to familiarize them with the task format without revealing the experimental purpose. Given their young age and thus limited attention span, the 45 experimental trials were divided into 3 blocks, with each block followed by a 3-5 minute break. The pilot test found that this break period allowed the participants to most comfortably continue the task.

3.4 Data analysis

The primary goal of the analysis was to test whether children's accuracy in selecting the correct picture is systematically influenced by the syntactic marking of telicity, and whether this sensitivity changes with age. To this end, I used logistic regression within the generalized linear mixed-effects modeling framework, implemented in R studio (version 4.5.1). This approach is ideally suited for the binary outcome (correct versus incorrect) and allows us to simultaneously account for the fixed effects of experimental conditions and the random variability introduced by individual participants and stimulus items.

For each experiment, I constructed a separate mixed-effects model. The model for experiment 1 included age group and condition (bare verb versus resultative verb compound) as fixed effects, along with their interaction. A significant interaction would indicate that the ability to use resultative complements as telicity markers develops across this age range. By-participant intercepts and by-item intercepts were included as random effects (Barr et al. 2013).

Apart from considering the main effect of age, condition, and their interaction, experiment 2 included particle type (*完* /wán/, *掉* /diào/, *到* /dào/, *好* /hǎo/, and *住* /zhù/) as an additional fixed factor. This operation tested for a three-way interaction (age×condition×particle type) to determine if the developmental profile differs across specific particles.

4. Results

4.1 Experiment 1

A mixed-effects logistic regression model tested the main effects of age and condition (bare versus resultative), along with their interaction. The model revealed a significant main effect of condition (β=-1.39, SE=0.67, z=-2.08, *p*=0.04). There was no significant main effect of age (age 4: β=0.11, SE=0.26, z=0.41, *p*=0.68; age 5: β=0.06, SE=0.26, z=0.22, *p*=0.83; age 6: β=0.23, SE=0.28, z=0.80, *p*=0.42). Crucially, a significant age×condition interaction emerged for the 6-year-old group (β=0.85, SE=0.37, z=2.31, *p*=0.02). This interaction was marginal for the 5-year-old group (β=0.65, SE=0.33, z=1.95, *p*=0.05) and non-significant for the 4-year-old group (β=0.30, SE=0.34, z=0.88, *p*=0.38). The observed effect sizes for the condition difference (bare versus telic) within each age group were moderate to large (age 3: d=0.70, age 4: d=0.54, age 5: d=0.38, age 6: d=0.28). These effect sizes indicate a developmentally decreasing magnitude of the condition difference.[5] See **Figure 8** for descriptive statistics and **Table 1** for full model statistics.

The random-effects structure revealed limited item-level variance (σ^2=0.75, SD=0.66) and small participant-level variance (σ^2=0.18, SD=0.42). Model comparison with random slopes for condition within participants did not improve fit ($\chi^2(2)$=1.24, *p*=0.54), indicating that individual differences in sensitivity to the condition manipulation were minimal.

Table 1. Full model statistics for experiment 1

	Estimate	**Std. Error**	***z*-value**	***p*-value**
(intercept)	3.03	0.50	6.08	<0.001
telic	-1.39	0.67	-2.08	0.04
age 4	0.11	0.26	0.41	0.68
age 5	0.06	0.26	0.22	0.83
age 6	0.23	0.28	0.80	0.42
age 4×condition	0.30	0.34	0.88	0.38
age 5×condition	0.65	0.33	1.95	0.05
age 6×condition	0.85	0.37	2.31	0.02

[5] If the the difficulty of the resultative compound condition was purely structural (i.e. the extra syllable/morpheme in the resultative compound condition taxing processing capacity), one would expect a stable disadvantage across all ages. The shrinking gap across age groups instead suggests progressive acquisition of the form-meaning mapping, not a fixed processing bottleneck.

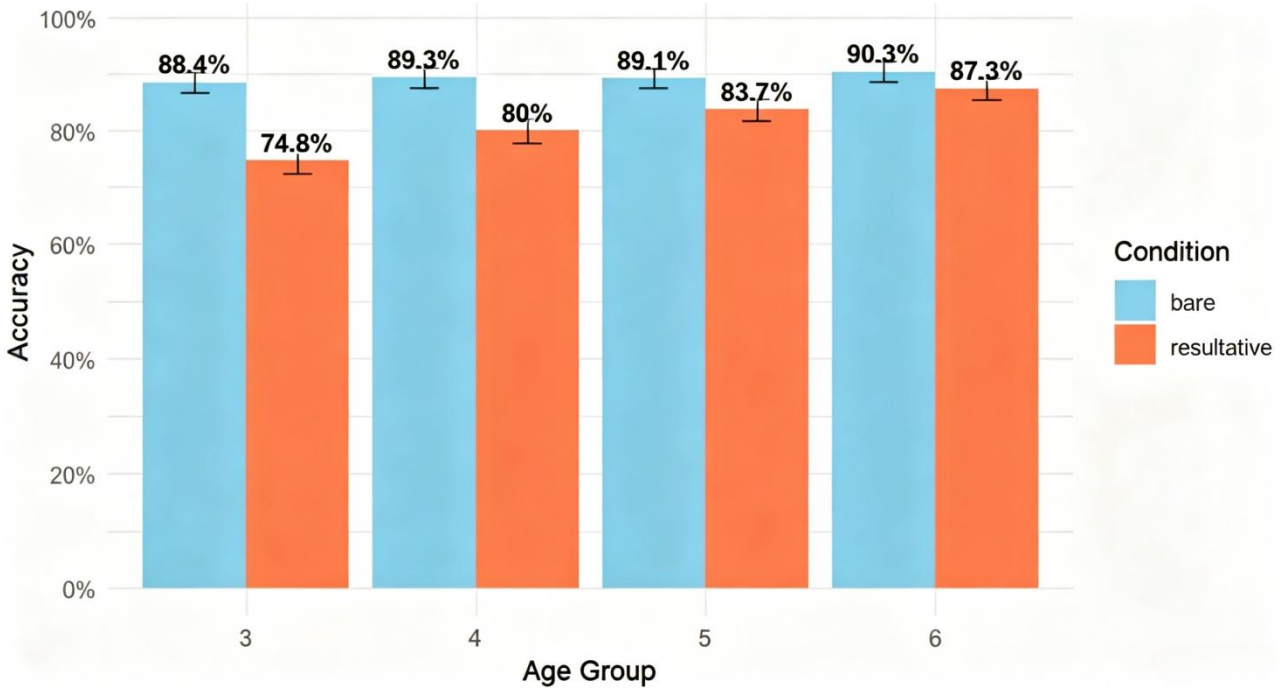


Figure 8. Accuracy by age and condition in experiment 1.

Note: Error bars represent standard error.

4.2 Experiment 2

A mixed-effects logistic regression model testing the main effects of age, condition (bare versus telic), and particle, along with their two-way and three-way interactions, was fitted to the data. The model revealed no significant main effect of condition (β=-0.41, SE=0.23, z=-1.78, *p*=0.18), no significant main effect of age (age 4: β=0.06, SE=0.31, z=0.18, *p*=0.86; age 5: β=0.15, SE=0.30, z=0.49, *p*=0.62; age 6: β=0.30, SE=0.29, z=1.03, *p*=0.31), and no significant main effect of particle (*掉* /diào/: β=-0.64, SE=0.47, z=-1.36, *p*=0.31; *到* /dào/: β=-0.13, SE=0.51, z=-0.25, *p*=0.80; *好* /hǎo/: β=-0.55, SE=0.48, z=-1.16, *p*=0.25; *住* /zhù/: β=-0.45, SE=0.48, z=-0.94, *p*=0.35). Critically, none of the interaction terms reached statistical significance: age×condition (age 4: condition: estimate=-0.05, SE=0.31, z=-0.15, *p*=0.88; age 5: condition: estimate=0.20, SE=0.31, z=0.65, *p*=0.52; age 6: condition: estimate=0.09, SE=0.30, z=0.29, *p*=0.77), age×particle, and the three-way age×condition×particle interaction. See **Figure 9** for descriptive statistics and **Table 2** for full model statistics.

Effect sizes for the key contrasts were relatively small. The condition effect (bare versus telic) within each age group yielded values of d=0.22 (age 3), 0.25 (age 4), 0.22 (age 5), and 0.18 (age 6). All fall near the conventional threshold for a small effect (d=0.20; Cohen, 1988). For pairwise comparisons among the five completive particles within the telic condition, the resulting effect sizes ranged from 0.11 to 0.25 (median≈0.17), also consistently within the small-effect range. This pattern of consistently small effect sizes across both critical contrasts corroborates the null findings from the inferential tests.

The random-effects structure revealed small participant-level variance (σ^2=0.45, SD=0.67) and minimal item-level variance (σ^2=0.04, SD=0.19). Model comparison with random slopes for condition within participants did not significantly improve fit ($\chi^2(2)=1.24$, p=0.54), indicating minimal individual differences in sensitivity to the condition manipulation. For the particle-specific model, the random effects showed small participant variance (σ^2=0.46, SD=0.68) but minimal item variance (σ^2=0.08, SD=0.13), suggesting that item-specific differences in this subset of data were negligible.

Table 2. Full model statistics for experiment 2

	Estimate	**Std. Error**	***z*-value**	***p*-value**
(intercept)	2.23	0.22	10.01	<0.001
telic	-0.41	0.23	-1.78	0.18
age 4	0.06	0.31	0.18	0.86
age 5	0.15	0.30	0.49	0.62
age 6	0.30	0.29	1.03	0.31
掉 /diào/	-0.64	0.47	-1.36	0.17
到 /dào/	-0.13	0.51	-0.25	0.80
好 / hǎo/	-0.55	0.48	-1.16	0.25
住 /zhù/	-0.45	0.48	-0.94	0.35

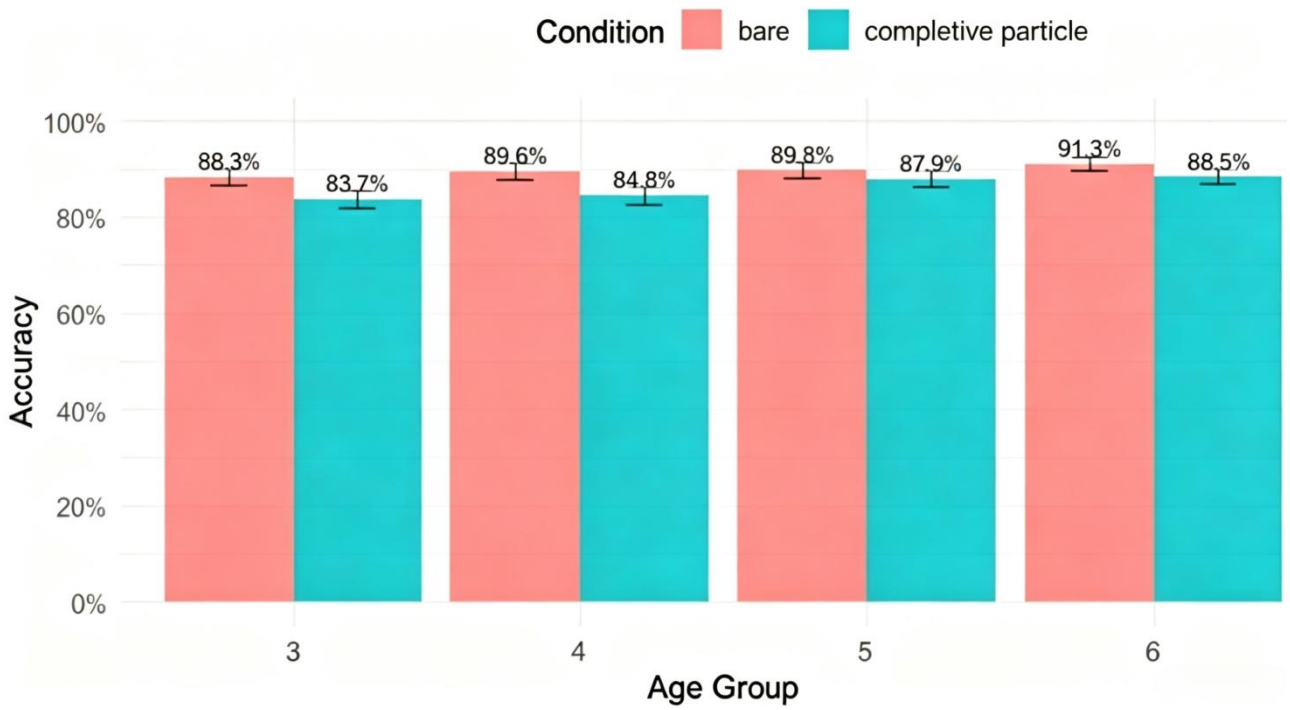


Figure 9. Accuracy by age and condition in experiment 2.

Note: Error bars represent standard error.

5. Discussion

The findings of experiment 1 present a nuanced picture of the developmental trajectory of telicity comprehension in Mandarin-speaking children, offering substantial evidence that informs the central theoretical debates outlined in the prior hypotheses. The core results demonstrate a significant main effect of condition, a developmental progression in accuracy for the resultative condition, and a significant age-by-condition interaction culminating at age 6. They address the research questions concerning children's successful recognition of the independent coding effect of result complements.

In dialogue with formal linguistics theory, the central acquisitional question under test is whether children, by the age of 3 and beyond, demonstrate sensitivity to verb-framed cues in a manner consistent with adult grammar. The positive evidence for this account is children's above-chance acceptance of telic interpretations for sentences containing resultative compounds without overt quantized objects, which was observed even at the youngest age tested and strengthened steadily. More critically, the systematic increase in accuracy from age 3 to age 6 in the resultative condition, contrasted with the uniformly high and stable performance in the bare-verb condition, indicates a

process of refinement. We could observe gradual sharpening and strengthening of the specific form-meaning mapping between the syntactic position of the resultative complement (Res) and the semantic feature [+telic]. Initially, children possess a partial mapping, as evidenced by their good-but-not-ceiling performance. This mapping becomes more precise, reliable, and automatized, converging on adult grammar where the resultative complement is a nearly obligatory signal of telic status. While the atelic baseline remains solid, the developmental pattern of selective refinement in the telic condition is consistent with predictions made by verb-framed approaches. It suggests that children are not simply learning individual verb-result pairs lexically but are abstracting a syntactic rule: the merger of a resultative element in ResP inherently makes the event telic. The absence of a similar developmental slope for bare verbs reinforces that this learning is specific to the syntactic encoding of telicity via ResP.

The interaction pattern, particularly its significance at age 6, further illuminates the nature of this syntactic learning. The lack of a main effect of age when conditions are collapsed underscores that children are not simply getting "better at the task" in a global sense (Calamia et al., 2012). Instead, the emerging sensitivity is highly selective, pertaining to the computation of this specific syntax-semantics interface. This finding challenges simpler maturational accounts and instead points to a prolonged period of syntactic consolidation. It aligns with the hypothesis that understanding these abstract, functional projections involved in aspectual composition is a hallmark of later language development, extending into the school-age year as metalinguistic abilities mature (Gonçalves et al., 2021; Melogno et al., 2022). The significant difference between 3-year-olds and both 5- and 6-year-olds in the resultative condition specifically highlights the period between 4 and 6 years as a critical phase for solidifying this aspect of syntax-semantics knowledge.

Turning to the contribution in developmental psychology, the results speak to the proposed interplay between domain-general event cognition and language-specific structural acquisition (Culbertson et al., 2013; Holtz, 2024). The study explored the extent to which children's early sensitivity to telic representations may influence their initial syntax-semantics organization. The data offer a complex but interpretable answer. The high and stable accuracy from age 3 in the bare verb condition is striking. To consistently select the "in-progress" picture for bare verbs, children must overcome the tendency to attend to salient event boundary in non-linguistic event perception (Baldwin et al., 2001; Loucks

& Meltzoff, 2013; Sonne et al., 2016). Their success suggests that by age 3, children are aware of using the absence of an overt resultative marker as a linguistic cue that constrains this general cognitive tendency. Thus, rather than eliminating an early telic bias, Mandarin-specific syntax-semantics knowledge guides children to interpret bare activity verbs as atelic events.

However, the concurrent lower and developing performance in the resultative condition reveals the other side of this language-cognition negotiation. It suggests that while children can inhibit a telic reading for bare verbs, fully exploiting the positive linguistic cue of a resultative complement requires additional learning. The resultative complement does not merely align with a general preference for a telic event interpretation; it specifies a particular kind of resultative state (e.g. open, broken). Children must learn which specific linguistic forms occupy this potent syntactic slot and reliably map them to both the abstract concept of telicity and sometimes to specific resultative states. The increasing accuracy with age reflects this learning process, where children gradually refine the mapping between resultative morphology and event structure. This idea aligns with a growing body of work in developmental psychology suggesting that language acquisition involves integrating domain-general perceptual sensitivity to event structure with the language-specific formal devices that encode aspectual distinctions in a given linguistic system (Jasmin, 2021; Saito et al., 2024; Smith, 2001; Werker, 2018).

Furthermore, the dissociation between the two conditions supports a dual-process model of event comprehension in language. The bare-verb condition may tap into a more automatic, heuristic process: from no overt result marker to likely not completed. This heuristic, once learned, is quickly stable. The resultative condition, conversely, requires a more controlled, compositional analysis: integrating the verb and the result complement, computing the resultative state, and matching it to a visual outcome. The development of executive functions (Yeager & Yeager, 2013), such as cognitive control for complex morpho-syntax-semantics integration, likely supports the improvement seen in the resultative condition across ages.

The results of experiment 2 present a contrasting and theoretically significant pattern to those of experiment 1. The absence of significant main effects for age or condition, the lack of a reliable age-by-condition interaction, and the null findings concerning particle type collectively indicate that

children between 3 and 6 years of age do not show a clear developmental progression in their ability to use completive particles as independent telicity markers in the context of this picture-selection task. While descriptive means show generally high and slightly variable accuracy across ages and particles, the statistical narrative is one of stability within the tested age range, posing a compelling pattern that requires careful theoretical unpacking.

This pattern of null results, particularly when juxtaposed with the clear developmental trajectory for resultative compounds in experiment 1, forces a crucial theoretical distinction between two syntactic mechanisms that are often grouped under the umbrella of verb-framed telicity. The findings suggest that completive particles and resultative verb compounds, while both base-generated in ResP under Ramchand's (2008) framework, present fundamentally different acquisition problems to young learners. A potential alternative is that Mandarin bare objects are covertly quantized. Under such a view, telicity would still ultimately depend on object-related quantization, albeit not overtly expressed. However, this account faces an empirical challenge from the present data. The experiments hold the nominal component constant while manipulating verb-framed morphology. If covert quantization were the primary source of telicity, it is not clear why children would show divergent developmental trajectories for resultative compounds and completive particles. One possibility is that covert quantization is contextually licensed rather than obligatory. Yet, if its activation is variable, learners must acquire when such quantization is available. The present findings suggest that children reliably treat completive particles as independent markers of telicity, whereas resultative compounds require more prolonged development. This asymmetry is thus not straightforwardly reducible to object-related factors alone.

Returning to the acquisitional pattern, the lack of a developmental slope for completive particles suggests they are acquired earlier and with less evident refinement in their telicity-marking function. Critically, this finding does not contradict the general observation that functional categories are challenging (Jordens, 2012; Shi, 2014). Rather, it underscores that the acquisition difficulty of a functional element depends in part on how transparent its semantic contribution is to learners. One plausible explanation is that completive particles are highly grammaticalized markers of telicity and carry a relatively unified functional load. Their core meaning is purely aspectual to signal an ended event without entailing a specific, visible change of state. This relatively abstract but uniform

meaning may be easier for children to map onto their existing sensitivity to event boundary than more varied and lexically specific meanings of resultative compounds. The terminative meaning aligns with children's early sensitivity to event boundary in non-linguistic event perception (Ji & Papafragou, 2024). The observed stability is thus consistent with the view that children bring this domain-general sensitivity to the task of acquiring language-specific aspectual markers.

In comparison, the protracted developmental trajectory for resultative compounds suggests that they impose greater demands on lexical and compositional integration, requiring children to map specific resultative states onto event structure. The divergence in acquisition trajectories is therefore more naturally captured if telicity can be encoded within the verb phrase via distinct morphosyntactic mechanisms, rather than being uniformly derived from object properties alone. Even if covert quantization is admitted as a possible component of the system, the present findings indicate an additional source of telicity (one independent of object quantification) is required to account for the observed developmental asymmetry.

Beyond semantic transparency, the stable performance of completive particles may also be facilitated by input frequency. Arguably, particles like 完 are more frequent in child-directed speech (MacWhinney, 2012), which likely promotes their early entrenchment as reliable, conventionalized signals of telicity. While this frequency-based account does not preclude the syntax-semantics role, it underscores the need to consider distributional properties of the input when explaining developmental trajectories in further research.

To account for cross-experimental implications for linguistic theory, the data suggest that from an acquisition standpoint, ResP may not be a monolithic syntactic category. While syntacticians may posit the same structural position for resultative complements and completive particles, children appear to treat them as distinct learning challenges. This supports the notion that the semantic content merged in Res---whether a full resultative state predicate or a light completive particle---critically impacts learnability. The acquisition data thus call for a model that can account for this difference in developmental timing. One possibility is that completive particles are initially acquired as functional heads associated directly with an abstract Asp(ect) head, separate from the predicate-oriented ResP that hosts resultatives. The steadily decent performance would then reflect the acquisition of a core

functional category of the clause (completive aspect), while the developmental understanding of resultatives reflects the complex integration of the lexical-conceptual structure (the resultative state) into the syntactic event frame.

In sum, the unexpected findings of experiment 2 are as informative as the positive findings of experiment 1. They indicate that the acquisition of telicity markers in Mandarin is not a unitary process. Completive particles, functioning as light, grammaticalized markers of telicity, appear to be integrated into children's event representations earlier and more stably than the more semantically heavy and specific resultative verb compounds. This dissociation indicates that the acquisition of verb-framed telicity is not monolithic and poses a challenge for a uniform syntactic analysis that treats resultative complements and completive particles as structurally identical. It suggests that semantic content (i.e. specific resultative state versus pure evental termination) critically modulates learnability. This insight enriches the parametric theory of telicity by demonstrating that even within a single language that favors verb-framed encoding, different syntactic instantiations of telicity follow distinct developmental timetables, likely shaped by their semantic transparency, functional load, and the extent to which they map onto children's early sensitivity to event boundary.

Finally, key methodological limitations must be noted. First, while it is necessary to prevent carryover effects and task fatigue, the use of two independent cohorts of children in two experiments means that the observed developmental dissociation between resultative compounds and completive particles could not be attributed solely to construction type. Inter-participant variability, though minimized by the stratified sampling, may introduce an alternative source of variance. Consequently, the claimed acquisitional trajectories may not be definitive. Future research employing a within-participant design with the same children completing both tasks would be better to confirm this cross-experimental contrast. Second, the current findings are only said to be consistent with and provide developmental motivation for a syntactic analysis that projects telicity from a verb-framed account; they do not definitively rule out an alternative lexical-constructionist account that also predicts telicity without object quantization. Future work can employ an artificial language learning paradigm that introduces novel verbs paired with telic markers (e.g. a nonsense resultative morpheme). If children readily extend the telic interpretation to these novel combinations, it would provide more direct evidence for an abstract, rule-based mapping over a lexical-constructionist account. A further

limitation concerns the scope of theoretical inference. The present study does not adjudicate between competing theories of adult grammar, as neither confirms developmental trajectories.

6. Conclusion

This study offers developmental evidence that constrains how telicity may be represented, suggesting that an analysis which allows telicity to be encoded through resultative morphology independently of object quantification may offer a nuanced account of the observed acquisition asymmetries. The divergent acquisition trajectories for resultative compounds and completive particles suggest that children are sensitive to fine-grained distinctions within the syntax of event structure. Such sensitivity interacts with non-linguistic attentional preferences for event endpoints documented in infancy. The results contribute to refining our understanding of the syntax-semantics interface, showing that the acquisition of aspectual distinctions is a protracted process of mapping language-specific forms onto a foundational cognitive architecture for event representation.

Supplementary Materials

See the full set of experimental stimuli via the link https://docs.google.com/document/d/1Zsk-huq3Ll9bjZua7PaP_SiVP6VET0yQOv0hGLE3MDk/edit?tab=t.0.

Interest Declaration

No conflict of interest is reported.

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